

temenos

Optimization of Test Scripts using AI-driven Analysis

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Report: Optimization of Test Scripts using AI-driven Analysis

Organization: Temenos

1 Executive Summary

In large-scale banking systems such as those developed by Temenos, automated testing plays a critical role in ensuring system reliability and regulatory compliance. However, the presence of over **6,00,000+ test scripts** has led to significant redundancy, inefficiencies and increased execution time.

This project focuses on **analyzing, identifying and optimizing duplicate or similar scripts** using data-driven and AI-assisted techniques by grouping scripts based on attributes such as **application, description and functionality**. The study aims to reduce duplication, improve execution efficiency and enhance maintainability.

The outcome of this project is a **structured optimization approach** that enables:

- Reduction in redundant scripts
- Improved test execution performance
- Better organization of test scenario

2 Challenges Faced by the Company

The large volume of scripts introduced several operational and technical challenges:

- **High Redundancy:** Multiple scripts performing similar or identical functions
- **Increased Execution Time:** Longer regression cycles due to unnecessary scripts
- **Maintenance Overhead:** Difficulty in updating and managing duplicated scripts
- **Inefficient Resource Utilization:** Increased computational and storage costs
- **Lack of Standardization:** Inconsistent descriptions and grouping of scripts
- **Limited Visibility:** Hard to identify overlap across applications and functions

3 Problem Statement

The banking system contains a massive repository of 6,00,000+ test scripts, many of which are duplicate or highly similar in nature.

The key problem is:

How can we efficiently identify, group and optimize redundant test scripts using data analysis and AI techniques to improve testing efficiency and reduce system overhead?

4 Solution Approach

Input

- Dataset with fields:
SCENARIO.ID, SCRIPT.ID, SCRIPT.GROUP, OLD.TPR, APPLICATION, TXN ID, FUNCTION, TPR.FILE.NAME , GENERATE.ERROR , PRODUCT.GROUP, DESCRIPTION
etc.
- Dataset shape: (6,22,665 , 12)

Process

The solution was implemented in multiple analytical layers:

1. Basic Understanding and Analysis

- Understanding of the dataset
- Grouping the columns based on their unique values
- Exploring methods to find duplicates

2. Data Cleaning & Preprocessing

- Removed inconsistencies in descriptions
- Standardized text data by trimming white spaces
- Created separate excels based on SCRIPT.GROUP

3. Initial Duplicate Identification

- Grouped scripts using **DESCRIPTION** field
- Identified exact duplicates using **Natural Language Processing (NLP)** based on SCRIPT.GROUP.

4. Enhanced Preprocessing for Duplicate Identification

- Combined **APPLICATION + DESCRIPTION** for filtering (APPLICATION used:)
- Used the above as **foreign key** in the main file, and combined SCRIPT.ID, DESCRIPTION, APPLICATION as delimited values
- Tried to build a **context** with the delimited descriptions using Agentic AI.

5. Duplicate identification

- Used **Agentic AI** to identify exact duplicates in the delimited file.

6. Clubbing and Optimization

- Highlighted the duplicates in the delimited file.
- Clubbed all the scenario id's under the same description

7. Dashboarding and Deployment

- Created an analytical dashboard page which contains data and visualizations of various results and key findings with the excels.
- Deployed the webpage in github-pages.

8. Analysis with other script groups such as SY.ACCU.DECU and FOREX after sorting strategy change.

Output

- Identified **duplicate and near-duplicate scripts**
- Reduced dataset complexity through grouping
- Created **insightful dashboard/report**
- Recommended **optimized script structure**

5 Implementation –Week-wise

Week	Activities	Outcome
Week 1 (7th Apr - 10th Apr)	- Understanding the dataset - Basic analysis and visualisations - Grouping excels based on SCRIPT.GROUP	A redundant group with a SCENARIO.ID "SCRIPT181000727230" and SCRIPT.ID "SCRIPT181000727230-003" called "TBO6" has been running while testing.
Week 2 (13th Apr - 17th Apr)	- Analysis made on TB01_MM.MONEY.MARKET only - Duplicates were found using NLP model	Initial threshold of 0.95% similarity failed to detect duplicates with mutually excluded terms such as "loan - deposit", "positive spread - negative spread", hence increased the threshold to 0.98% and added a business logic which handles mutually exclusive terms.
	Duplicates were found using the NLP model with increased threshold and an added business logic.	The SCRIPT.GROUP based analysis was not enough as it did not consider a scenario entirely. Changed the strategy to just take the scenario id from MM.MONEY.MARKET_TB01 and compare it with the main dump.
Week 3 (20th Apr - 24th Apr)	The SCENARIO.ID in MM.MONEY.MARKET_TB01 was extracted and was used for filtering Scenarios in the main dump which had APPLICATION	Highlighted the duplicates in the delimited file.

	== MM.MONEY.MARKET and SCRIPT.GROUP == TB01. • These SCENARIO.ID's were used in the SCRIPTS dump, and new delimited files with all the necessary values were created. • The delimited file was put into an Agentic AI to find out duplicates based on DESCRIPTION only. • The same was done for APPLICATION == SY.ACCU.DECU and SCRIPT.GROUP == TB01.	• Clubbed all the scenario id's under the same description • Created an analytical dashboard page which contains data and visualizations of various results and key findings with the excels. • Deployed the webpage in github-pages. • FEEDBACK AFTER REVIEW: FUNCTION and duplicated SCRIPT.IDs in the same SCENARIO .ID must be considered. A new field "RF.FIELD.NAME" can be introduced for better duplicate detection.
Week 4 (27th Apr - 4th May)	• Detailed documentation with necessary evidences and screenshots • Cleaning of the RF.FILE.NAME column by understanding the special characters present and filling the garbage values with a unique "I". • Duplicate analysis done using RF.FIELD.NAME	• Duplicate report was derived keeping Script id's of TB01 and MM.MONEY.MARKETc including the RF field.
Week 5 (12th May -15th May)	• Delimited files for the SCENARIO's belonging to TB01 and FOREX were created.	• Duplicate report was derived. • The sorting of the script groups were different. So, the sorting strategy was changed.

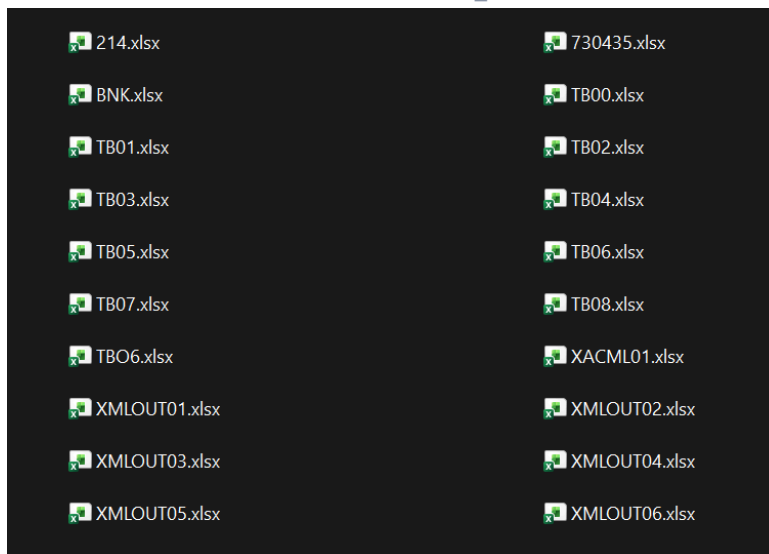
Week 6 (18th May-22nd May)	• The sorting strategy was changed. Firstly, the script groups of a particular scenario were sorted in the following manner. • TB0x-START, TB0x, TB0x-End. • Then, if the TBOx-y is the same, they were sorted according to TPR. • The delimited excel was properly created. • The flow column contains all the important business logic keywords.	• The delimited file was analyzed and a duplicated report was derived. • A flow column was added for future duplicate analysis. • The above was also derived for the application FOREX
Week 7 (25th May-29th May)	• Duplication verification and documenting.	

6 Results and Findings

6.1 Week 1 - Dataset Understanding and Presence of a Redundant SCRIPT.GROUP

SCENARIO.ID	SCRIPT.ID	SCRIPT	OLD	APPLICATION	TXN ID	FU	TPR	U	PRO	GEN	DESCRIPTION
SCRPT06029020187	SCRPT06029020187	TB02	11	FACILITY	SCRPT06029020187	I	2.tpr		CORPCNO		Script to change Fee rates TEST BASE MERGE
SCRPT06043051077	SCRPT06043051077	TB00	1	CUSTOMER	802014	I	1.tpr		TREASINO		TEST BASE MERGER BASE DATA RECORD FO
SCRPT08141056047	SCRPT08141056047	TB01-STAF	2	ACCOUNT	1E+10	I	2.tpr		CORPCNO		BL Collection Multiple BR Remove BR attach
SCRPT08240050113	SCRPT08240050113	TB01-STAF	3	ACCOUNT	1E+10	I	3.tpr		TREASINO		TEST BASE MERGER BASE DATA RECORD FO
SCRPT06029020187	SCRPT06029020187	TB02	12	SL.RATES.PART	SCRPT06029020187	IA	3.tpr		CORPORATE		Script to change fee rates TEST BASE MERGE

6.1.1 Dataset SCRIPT_FINAL



6.1.2 Script Group based filtering

SCENARIO.ID	SCRIPT.ID	SCRIPT.GROUP	OLD.TPR	PPUCATIO	TXN ID	FUNCTION	R.FILE.NAM	SER.NAM	DUCT.GRO	ATE.	DESCRIPTION
SCRPT181000727	SCRPT181000727230-003	TB06-END	3	CUSTOMER	727272	I	5.tpr		SECURITI	NO	EN 2207136 EQUITY INSTRUMENT

6.1.3 Redundant Group "TBO6"

SCENARIO.ID: SCRPT181000727230

SCRIPT.ID: SCRPT181000727230-003

SCRIPT.GROUP: TBO6 - END

6.2 Week 2 - Analysis in TB01-MM.MONEY.MARKET USING NLP

Dataset:

SCENARIO.ID	SCRIPT.ID	RPT.GRO	OLD.TPR	APPLICATION	TXN ID	INCTIC	FILE.NR	PRODUCT.GRO	RATE	DESCRIPTION
SCRPT0604305	SCRPT060	TB01	2	MM.MONEY.MARKET		IA	2.tpr	TREASURY	NO	Deposit Dea
SCRPT1716308	SCRPT171	TB01	3	MM.MONEY.MARKET		I	1.tpr	TREASURY	NO	Input floating
SCRPT1712808	SCRPT171	TB01	4	MM.MONEY.MARKET		IA	1.tpr	TREASURY	NO	To create a M
SCRPT0809105	SCRPT080	TB01	2	MM.MONEY.MARKET		IA	2.tpr	TREASURY	NO	HD-Inputting
SCRPT1521101	SCRPT152	TB01	6	MM.MONEY.MARKET		I	1.tpr	BANKING.FRNO		Create MM C

6.2.1 Dataset MM.MONEY.MARKET_TB01

SCENARIO	SCRIPT.ID	SCRIPT	OLD.T	APPLICATI	TXN ID	FUNCTPR	FILE USE	PROD	GEN	DESCRIPTION	BASE_GRC	DUPLICATE_GROUP
SCRPT171	SCRPT171	TB01	3	MM.MONEY.MARKE	I	1.tpr		TREAS	NO	Input floating nc	TB01	Group_1
SCRPT171	SCRPT171	TB01	3	MM.MONEY.MARKE	I	1.tpr		TREAS	NO	Input floating nc	TB01	Group_1
SCRPT171	SCRPT171	TB01	3	MM.MONEY.MARKE	I	1.tpr		TREAS	NO	Input floating nc	TB01	Group_1
SCRPT171	SCRPT171	TB01	3	MM.MONEY.MARKE	I	1.tpr		TREAS	NO	Input floating nc	TB01	Group_1
SCRPT171	SCRPT171	TB01	3	MM.MONEY.MARKE	I	1.tpr		TREAS	NO	Input floating nc	TB01	Group_1

6.2.2 Dataset with duplicates in MM.MONEY.MARKET_TB01

QSI - MM.INTEREST Inputting an MM Call Deposit with Negative Interest. EN - 204404	Group_101
QSI - MM.INTEREST Inputting an MM Deposit Deal with Negative Interest. EN - 204404	Group_101

6.2.3 Descriptions with mutually excluded terms were grouped together

- A “**business logic**” was added to ensure that terms having mutually exclusive words should not be grouped as duplicates.

Rule B: Mutually Exclusive Banking Terms

```
exclusive_lists = [
    ['loan', 'deposit'],
    ['accrual param as first', 'accrual param as last', 'accrual param as both', 'without accrual param'],
    ['positive spread', 'negative spread']
]
```

- Increase of similarity threshold was also performed.
- Needed a change in strategy because these duplicates were only based on TB01 and MM.MONEY.MARKET.
- **Strategy change:** Consider the SCENARIO.ID present in MM.MONEY.MARKET_TB01 file and use it as a key to get the data from the SCRIPT_FINAL file.

6.3 Week 3 - Analysis in the main SCRIPT file

SCENARIO.ID	SCRIPT.ID	DESCRIPTION	FUNCTION	APPLICATION
SCRPT06023051179	001 002 003	TEST BASE MERGER. B IA IA		CUSTOMER MM.MONEY.MARKET MM.MONEY.MARKET
SCRPT06023051181	001 002	Deposit Deal with FixeIA IA		MM.MONEY.MARKET MM.MONEY.MARKET
SCRPT06023051183	001 002 003	TEST BASE MERGER FO IA IA		CUSTOMER MM.MONEY.MARKET MM.MONEY.MARKET
SCRPT06023051185	001 002 003	TEST BASE MERGER FO IA IA		CUSTOMER MM.MONEY.MARKET MM.MONEY.MARKET

6.3.1 Delimited file of all the scenario ids

SCRPT113140800002	105 205 110 200	Creating a customer for the Defect # 303909. "System throws error message "FINAL MATURITY DATE LESS THAN MATURITY" Preclosure of the MM contract MM0935701176 on day 3. Defect # 303909. "System throws an error message " FINAL MATURITY DATE LESS THAN MATURITY" Creating an Account for the Defect # 303909. "System throws error message "FINAL MATURITY DATE LESS THAN MATURITY" Input of an MM CALL deposit contract with I function alone for the Defect # 303909. "System throws error message "FINAL MATURITY DATE LESS THAN MATURITY"	I IA I IA
SCRPT113140800003	105 205 110 200	Creating a customer for the Defect # 303909. "System throws error message "FINAL MATURITY DATE LESS THAN MATURITY" Preclosure of the MM contract MM0935701176 on day 3. Defect # 303909. "System throws an error message " FINAL MATURITY DATE LESS THAN MATURITY" Creating an Account for the Defect # 303909. "System throws error message "FINAL MATURITY DATE LESS THAN MATURITY" Input of an MM CALL deposit contract with I function alone for the Defect # 303909. "System throws error message "FINAL MATURITY DATE LESS THAN MATURITY"	I IA I IA

6.3.2 Duplicated Descriptions are highlighted

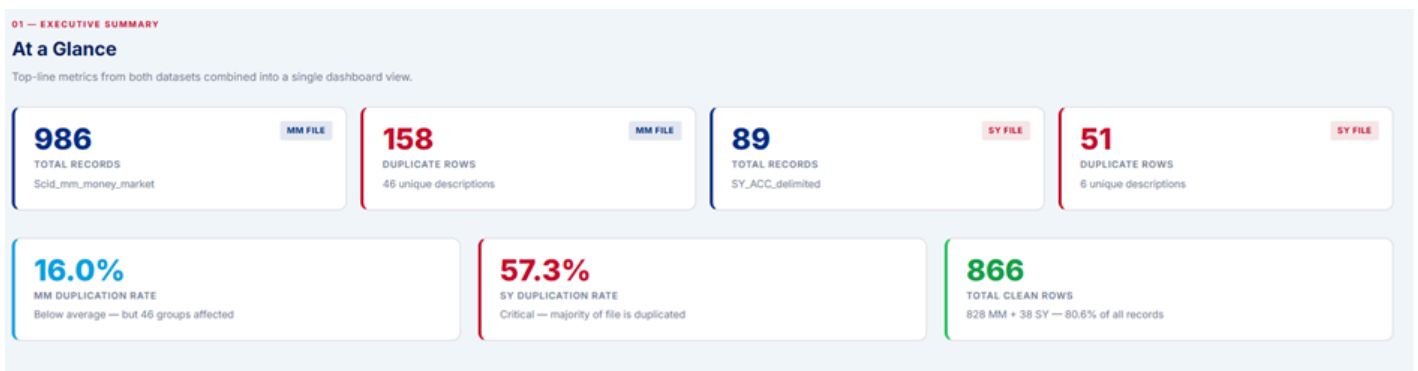
#	DESCRIPTION	Duplicate Count	SCENARIO IDs
1	AM-Back Value Money Market Transaction To view in AM Front-End TEST BASE MERGER	10	SCRPT161000613126, SCRPT161000613113, SCRPT161000613112, SCRPT161000613111, SCRPT161000613110, SCRPT161000613109, SCRPT161000613124, SCRPT161000613125, SCRPT161000613127, SCRPT161000613128
2	AM-Forward Value Money Market Transaction To view in AM Front-End TEST BASE MERGER	10	SCRPT161000613119, SCRPT161000613123, SCRPT161000613138, SCRPT161000613122, SCRPT161000613120, SCRPT161000613121, SCRPT161000613137, SCRPT161000613136, SCRPT161000613135, SCRPT161000613134

6.3.3 Duplicated Descriptions mapped with their SCENARIO.IDs

Dashboard Link:

<https://sanjanav2102.github.io/Test-Script-Optimization-and-Duplicate-Analysis-Dashboard/>

- **SY.ACCU.DECU File is Most Critical** with a **57.3% duplication rate**, more than half of SY_ACC contains repeated descriptions. **"EN 1426890"** alone appears 21 times, an entire test group bulk-copied without updating the description field. Immediate cleanup is required.
- **MM.MONEY.MARKET File Has the Widest Spread** with **46 unique duplicate groups**, the MM file's duplication is far broader. The top 3 descriptions each appear **10 times** and all relate to TEST BASE MERGER scenarios, suggesting bulk copy-paste during test creation cycles.
- **Naming Inconsistency Hides More Duplicates** **"EN 1388099"** (space) and **"EN-1388099"** (hyphen) reference the same enhancement but are stored as different values, creating 11 hidden duplicates. Standardization is required here.



6.4 Week 4 - Analysis in the main SCRIPT file + RF.FILE.NAME

SCENARIO	SCRIPT.ID	SCRIPT.GR	OLD.TPR	APPLICATION	TXN ID	FUN	TPR.FIL	U	PROD	GENER	DESCRIPTION	RF.FILE.NAME
SCRPT06C	SCRPT060	TB02	11	FACILITY	SCRPT060	I	2.tpr		CORP	NO	Script to chan	FACILITY,S
SCRPT06C	SCRPT060	TB00	1	CUSTOMER	802014	I	1.tpr		TREAS	NO	TEST BASE ME	MM.MON
SCRPT081	SCRPT081	TB01-STAF	2	ACCOUNT	1.01E+10	I	2.tpr		CORP	NO	BL Collection	BL.REGIST
SCRPT082	SCRPT082	TB01-STAF	3	ACCOUNT	1.01E+10	I	3.tpr		TREAS	NO	TEST BASE ME	ND.DEALS

6.4.1 Updated SCRIPTS with RF.FILE.NAME column

Metric	Value
Total rows in CSV (SCENARIO's of records having MM.MONEY.MARKET AND TB01)	989
Rows with duplicate DESCRIPTION (from XLSX)	980
TRUE Duplicate rows (desc + RF same)	125
Unique TRUE duplicate groups	36
Most frequent single group	10
Least frequent group (min count)	2
Average duplicates per group	3.5

6.4.2 Updated SCRIPTS with RF.FILE.NAME column

Out of 989 total records, 125 records were identified as true duplicates, resulting in an overall duplication rate of approximately **12.64%**.

6.5 Week 5 - Analysis in the main SCRIPT file + RF.FILE.NAME for FOREX

Forex TB01 — Duplicate Analysis Statistics

Total rows in file	194
Total duplicate rows	11
Clean (non-duplicate) rows	183
Duplication rate (%)	5.7
Unique duplicate groups	4
Max occurrences (single group)	4
Duplicate rule	DESCRIPTION + FUNCTION + RF.FILE.NAME all identical

6.6 Week 6-7 - Analysis after sorting strategy change

Excel link:

https://docs.google.com/spreadsheets/d/10J7hyG-BczQNdibMFhPWWvgWF_2Be9BC/edit?usp=sharing&ouid=105424369009839867770&rtpof=true&sd=true

1. Overall Regression Suite Context		
Total Regression Scripts (Enterprise)	623461	Full suite spanning all applications (623,461 scripts × 13 columns)
Applications Analysed	2	FOREX and MM.MONEY.MARKET only
Script Group Filter	TB01	Only TB01-tagged scenarios analysed for duplication

2. FOREX Application – TB01 Duplicate Analysis		
Total FOREX Rows (all groups)	845	Total scenarios in FOREX file
Duplicate Rows Found	76	Rows matching on DESCRIPTION + FUNCTION + RF.FILE.NAME
Unique Duplicate Groups	31	Distinct duplicate scenario clusters
Removable Duplicates	45	Keeping 1 per group; remainder can be removed
Deduplication Rate (FOREX TB01)	6.9%	45 out of 656 TB01 rows are redundant

3. MM.MONEY.MARKET Application – TB01 Duplicate Analysis

Total MM Rows (all groups)	997	Total scenarios in MM file
Duplicate Rows Found	113	Rows matching on DESCRIPTION + FUNCTION + RF.FILE.NAME
Unique Duplicate Groups	30	Distinct duplicate scenario clusters
Removable Duplicates	83	Keeping 1 per group; remainder can be removed
Deduplication Rate (MM TB01)	10.1%	83 out of 820 TB01 rows are redundant

4. Combined (FOREX + MM) TB01 Duplicate Summary

Combined TB01 Scenarios	1476	Across both FOREX and MM applications
Combined Duplicate Rows	189	Total rows flagged as duplicates
Combined Duplicate Groups	61	Unique duplicate clusters across both apps (30+31)
Total Removable Duplicates	128	Scripts that can be safely eliminated
Combined Deduplication Rate	8.7%	Of TB01 scenarios in these 2 apps

5. Efficiency Projection – If Deduplication Applied Across Full Suite		
Full Suite Total Scripts	623461	Total enterprise regression scripts
Observed Duplication Rate (2 apps)	8.7%	Based on 128 removable from 1476 TB01 scenarios
Projected Removable Scripts (if extrapolated)	~54,067	Indicative only. Actual savings depend on duplication rates in all other applications.
Projected Optimised Suite Size	~569,394	Estimated suite size after deduplication across all apps (if rate holds)
Estimated Execution Time Reduction	~9%	Fewer scripts executed per regression cycle → faster feedback loops

CONCLUSION

The total regression suite has **623,461 scripts**. With an ~8.5% observed duplication rate across the 2 applications analysed:

- **Projected removable scripts suite-wide:** ~54,000 (if the rate holds across all apps)
- **Estimated execution time reduction:** ~8–10% per regression run
- More concretely — even the 128 scripts already identified mean those test slots, compute cycles, and report noise can be eliminated immediately. The real gain comes from applying the same **DESCRIPTION + FUNCTION + RF.FILE.NAME** scan to all other applications in the suite, where the duplication pattern is very likely to repeat.

The **FLOW** column is designed to act as a compact fingerprint going forward — identical flows with different Scenario IDs are a strong signal of duplication worth reviewing.